MILITARY SPECIFICATION

CALCIUM THIOGLYCOLLATE, TRIHYDRATE, CREAM

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers Calcium Thioglycollate, Trihydrate, Cream (see 6.2.)

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-C-186 - Containers, Packaging and Packing for Drugs, Chemicals and Pharmaceuticals.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Personnel Support Center, Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6505

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

U. S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE,
FOOD AND DRUG ADMINISTRATION (FDA)

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Food and Drug Administration, Washington, DC 20204.)

U. S. PHARMACOPEIAL CONVENTION, INC.

The United States Pharmacopeia (USP).

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

The National Formulary (NF).

(Application for copies should be addressed to the Mack Publishing Company. Easton, PA 18042.)

3. REQUIREMENTS

3.1 Material. Shall be a smooth, nonirritating cream free from grit and shal contain calcium thioglycollate trihydrate, 5.0 percent, within the designated ass limits, in a suitable water soluble, emollient base containing a suitable preservative. The cream shall be suitable for use as a depilatory.

3.1.1 Assay. The cream shall assay to contain not less than 4.5 percent and not more than 5.5 percent calcium thioglycollate, when assayed as specified in 4.7.1.

3.1.2 pH. The pH of a 10 percent slurry of the cream in purified water shall be not less than 10.0 and not more than 12.0 when determined potentiometrically using the USP method and a high-alkali electrode.

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3.2 Ingredients.

3.2.1 Calcium thioglycollate, trihydrate.

3.2.1.1 Reaction. Solutions shall be alkaline.

3.2.1.2 Identification. Shall respond to the flame test for calcium. Shall have an odor of the thioglycollic acid.

3.2.1.3 Assay. Shall contain not less than 94.0 percent calcium thioglycollate, trihydrate, when assayed as specified in 4.7.2.

3.3 Other ingredients. All other ingredients used in the manufacture of the cream shall be in accordance with the tests, standards, and requirements of the USP or NF, including any supplements or revisions thereto; or, if not included in either of these compendia, the ingredients shall be of the highest pharmaceutical grade.

3.4 Depilatory action test. The treated area shall be completely devoid of hair, with no short hair or stubble remaining, when tested as specified in 4.7.3.

3.4.1 Irritation. The area of the forearm used for the depilatory action test shall show no signs of irritation when examined immediately after treatment and for a period of 24 hours thereafter.

3.5 Expiration dating period. The expiration dating period shall be not less than 30 months.

3.5.1 Delivery. Not more than 3 months of the expiration dating period shall have elapsed at the time of delivery to the Government.

3.6 Workmanship. The material and its containers shall be free from defects which detract from their appearance or impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and, (i) as to any expiration dated item, for the expiration dating period specified by the contractor for such item, or such longer period as may be required by regulation of any federal agency; or (ii) as to non-dated items, for not less than 3 years after delivery of the item to the Government.

4.1.2 No company supplying any ingredient(s) to the contractor will be considered an acceptable facility for the performance of any inspection requirements specified herein.

4.2 Lot. For purposes of this specification, a lot, batch, or control is that single, uniform, and homogeneous quantity of cream produced from one formulation, subjected to the same compounding and manufacturing operation, and filled into final containers.

4.3 Sampling. Sampling shall be conducted in accordance with MIL-STD-105.

TABLE I. Sampling.

For visual examination	Inspection level	AQL (percent defective)	Unit of product
Major	S-2	1.0	Filled immediate container
Minor	II	2.5	Filled immediate container

4.3.1 Raw material and end item testing. Noncompliance with any of the following is cause for rejection.

TABLE II. Criteria for raw material and end item testing.

Characteristic	Requirement paragraph	Inspection level 1/	Minimum sample
Assay	3.1.1	S-2	5
pH	3.1.2	S-2	5
Depilatory action test	3.3	S-2	5
Irritation Ingredient 2/	3.3.1	-	-
Calcium Thioglycollate Trihydrate	3.2.1	-	-

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TABLE II. Criteria for raw material and end item testing.(cont'd.)

/ Where possible, the sample shall be used for two or more tests.
/ Sufficient quantities of each of the raw materials shall be selected from each lot to perform tests indicated in applicable test methods. Failure of any sample to comply with test requirements shall be cause for rejection of the lot represented.

4.4 Examination. The finished cream shall be examined to determine compliance with all requirements of this specification. Nonconformance will be permitted to the extent indicated in 4.3 and 4.3.1.

4.4.1 Classification of defects. Examination will be conducted in accordance with the following classification of defects.

TABLE III. Classification of defects.

ategories	Defects 1/
ajor	
101	Cream not free from grit.
102	Cream has crystal formation.
103	Cream not homogeneous.
104	Cream not smooth.
inor	
201	Exterior of immediate container is not free from contents or foreign matter.
202	Label not free from defacing marks or tears.

/ Inspection is not restricted to the classified possible defects listed above.

4.5 Packaging inspection. The inspection of the intermediate package, acking and marking for shipment and storage shall be in accordance with quality ssurance provisions of the applicable container specification and the marking equirements of MIL-STD-129.

4.6 Labeling inspection. Sampling shall be in accordance with MIL-STD-105 nd the unit of product shall be the unit of issue. The inspection lot shall e the number of units of issue of each batch, lot, control number, etc., offered or inspection at one time. The inspection level for labeling shall be S-2 nd the acceptance number shall be zero.

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4.7 Tests.

4.7.1 Assay for calcium thioglycollate, trihydrate, in the finished cream.

Accurately weigh about 1 gram of the finished cream into a flask, add 50 ml of purified water and 5 ml of Hydrochloric Acid, USP, with constant stirring. Wash the glass rod with a few ml of purified water and add to the flask. Add porcelain boiling chips, and carefully boil for approximately 7 minutes, preventing boiling over. Cool to room temperature, add 1 ml of starch T S and titrate with 0.025N iodine to a purple endpoint which persists for 1 minute. Perform a concomitant blank.

Calculation:

$$\frac{(\text{ml } I_2 \text{ Sample} - \text{ml } I_2 \text{ Blank}) \times \text{normality } I_2 \times 184.2 \times 100}{\text{Weight of Sample (mg)}} = \text{Percent Calcium Thioglycollate, Trihydrate}$$

4.7.2 Assay for calcium thioglycollate, trihydrate (ingredient).

Accurately weigh about 600 to 700 mg of calcium thioglycollate, trihydrate, into a 250 ml iodine flask. Dissolve in 100 ml of purified water. Add 5 ml Hydrochloric Acid, USP, and 3 grams of KI. Titrate with 0.1N KIO_3 using starch T S as the indicator. Each ml of 0.1N KIO_3 is equivalent to 18.423 mg of Calcium Thioglycollate, Trihydrate.

4.7.3 Depilatory action test.

Spread a layer of 1 to 2 mm of the finished cream, aged at least 7 days from the date of manufacture, onto a hairy area of the forearm. Exactly 10 minutes later, remove the cream, wash and dry, and examine.

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with all applicable requirements of PPP-C-186, and as specified herein.

5.2 Packaging. Packaging shall be level A or C as specified (see 6.1).

5.2.1 Unit of issue. One tube containing 100 grams, as specified, constitutes one unit of issue.

5.2.2 Level A.

5.2.2.1 Immediate container. The immediate container shall be a commercially available collapsible tube considered acceptable by the Food and Drug Administration (FDA) for the product contained therein and shall be of such compatibility that neither the contents nor the container are altered in any way by each other.

5.2.2.2 Unit package. At the option of the contractor, each unit may be packaged in a suitable container.

5.2.2.3 Intermediate package. Twelve (12) units of issue shall be packaged as specified in 5.2.13 of PPP-C-186.

5.2.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.3 Packing. Packing shall be level A, B or C, as specified (see 6.1).

5.3.1 Level A. One hundred forty-four (144) units of issue shall be packed as specified in 5.4.4 of PPP-C-186.

5.3.2 Level B. One hundred forty-four (144) units of issue shall be packed as specified in 5.4.3.5.1 of PPP-C-186.

5.3.3 Level C. The packaged commodity shall be packed in shipping containers that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carrier's rules and regulations.

5.4 Marking.

5.4.1 Labeling. Labeling shall be in accordance with the requirements of the Federal Food, Drug and Cosmetic Act and regulations promulgated thereunder, and shall include the information required below:

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5.4.1.1 Immediate container: Each immediate container label shall bear the following information:

(a) label information in accordance with commercial practice.
(b) the expiration date.

5.4.1.2 Unit package. Each unit package label shall bear the following information:

(a) label information in accordance with commercial practice.
(b) the National Stock Number.

NOTES: (1) The National Stock Number need not be preceded by any prefix.
(2) Strip labeling may be used.
(3) Label adhesion test shall not apply.
(4) If a unit package is not supplied, the National Stock Number shall appear on the immediate container label.

5.4.2 Intermediate package. Intermediate package shall be marked as specified in 5.5.3 of PPP-C-186, Type I shelf life markings as specified in MIL-STD-129 shall be shown.

5.4.3 Exterior container. Exterior container shall be marked as specified in 5.5.4 of PPP-C-186. Type I shelf life marking as specified in MIL-STD-129 shall be shown.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of specification.
(b) National Stock Number.
(c) Selection of applicable levels of packaging and packing (see 5.2 and 5.3).

6.2 This specification covers the following item :

National Stock Number	Item Identification
6505-00-926-8987	CALCIUM THIOGLYCOLLATE, TRIHYDRATE, CREAM, Surgical Depilatory, 5%, 100 Grams

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activity:
DoD - MB

Preparing activity:
DLA-DM

Project No. 6505-1786

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-C-37644 - CALCIUM THIOGLYCOLLATE, TRIHYDRATE, CREAM

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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